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TCN-QRNN model for short term energy consumption forecasting with increased accuracy and optimized computational efficiency

Lesia Mochurad✉ & Roman Levkovych

In the context of the growing volume and complexity of data, traditional methods of energy consumption forecasting, such as Recurrent Neural Networks (RNN), face computational complexity issues that limit their real-time application. This also complicates the effective management of energy systems. In this work, a new model is proposed that combines the advantages of Temporal Convolutional Networks (TCN) and Quasi-Recurrent Neural Networks (QRNN) for energy consumption forecasting. TCN allows for effective processing of long time series, capturing essential temporal dependencies. Meanwhile, QRNN reduces computational costs through parallelization of operations and an optimized architecture. The effectiveness of the proposed model has been assessed in comparison with traditional methods such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks, as well as other convolutional approaches. Experimental results show that the proposed TCN-QRNN model outperforms traditional methods by 40% in accuracy compared to LSTM and by 8% in terms of metrics such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) compared to TCN-LSTM, while reducing data processing time by 30%. Additionally, the model has a significantly smaller number of parameters than LSTM and GRU, making it suitable for environments with limited computational resources. The proposed model ensures a high level of energy consumption forecasting accuracy while significantly reducing processing time, making it promising for use in real-world energy systems.

Keywords Time series forecasting, Neural network optimization, Model efficiency, Parallel processing, Resource-constrained environments

Abbreviations

GRU	Gated Recurrent Unit
LSTM	Long Short-Term Memory
TCN	Temporal Convolutional Network
QRNN	Quasi-Recurrent Neural Network
TCN-GRU	Temporal Convolutional Network – Gated Recurrent Unit
TCN-LSTM	Temporal Convolutional Network – Long Short-Term Memory
TCN-QRNN	Temporal Convolutional Network – Quasi-Recurrent Neural Network
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MSE	Mean Squared Error
RMSE	Root Mean Squared Error
ADF	Augmented Dickey-Fuller test
KPSS	Kwiatkowski–Phillips–Schmidt–Shin test
SiLU	Sigmoid Linear Unit
ReLU	Rectified Linear Unit

Energy consumption forecasting is an important problem for different industries, especially in an energy industry where forecasting accuracy directly affects the efficiency of energy system management[1]. The need for precise and efficient energy consumption forecasting has grown significantly due to the increasing reliance

Lviv Polytechnic National University, Lviv 79013, Ukraine. ✉email: lesia.i.mochurad@lpnu.ua

on data in modern energy systems[2], particularly for microgrids and virtual power plants that require careful resource planning. Overcoming traditional forecasting methods' limitations requires innovative approaches combining computational efficiency with high prediction accuracy. Traditional methods sometimes fail to deliver the required level of precision and efficiency[3,4] given the growing complexity and volume of real-time data. Particularly, forecasting methods based on classical approaches do not take into account the complexity and nonlinearity of interconnections in time series that may lead to forecasting errors[5].

In this context, large volumes of historical power consumption data necessitate advanced forecasting methods that combine the strengths of various architectures. Deep learning models have recently gained attention for time series prediction tasks[6,7]. Although recurrent neural networks (RNNs) are traditionally used for such problems, their sequential nature limits computational efficiency. In contrast, convolutional operations enable parallel processing and effective extraction of temporal features, enhancing both speed and predictive performance. TCN[8] and QRNN[9] represent two such approaches that incorporate convolutional layers in different ways. While TCNs offer efficient modeling of long-range dependencies, QRNNs reduce computational overhead by combining convolution with gated mechanisms. This study proposes a lightweight hybrid model that leverages the complementary advantages of TCN and QRNN for power consumption forecasting. The proposed approach ensures high accuracy while significantly reducing training time and resource requirements when working with large-scale time series data.

In addition, validating the proposed model under real-world conditions is essential to ensure its reliability for battery-supported microgrids, which must respond rapidly to changing weather patterns, fluctuating energy demand, and variable electricity prices. In such distributed systems, where real-time forecasting of energy consumption and generation is critical, fast and adaptive models are required to dynamically adjust to fluctuations in solar irradiance, wind speed, and local load. These variations have a significant impact on the operational efficiency of microgrids and the performance of battery storage systems.

As highlighted in[10], robust model validation plays a vital role in scenarios where forecasting errors may jeopardize system stability. To address this, the Temporal Logic of Actions (TLA+) and PlusCal formalisms were employed as part of the verification process, enabling formal specification and validation of functional system properties. An effective example is the evaluation of energy management strategies in distributed networks, where alternative validation approaches—such as depth-first and breadth-first search algorithms – were analyzed under multithreaded execution, taking into account resource constraints in edge computing environments.

In the scope of the study, we evaluate the performance of the proposed model in comparison with traditional methods, such as LSTM and GRU, and other convolutional approaches used to forecast time series. The primary focus is improving accuracy and reducing processing time[10,11], which are critical factors for applications in the energy industry.

The main contributions can be summarized as follows:

1. A new approach is proposed that, for the first time, combines TCN and QRNN for energy consumption forecasting. TCN is employed to provide an in-depth analysis of temporal characteristics through multi-layer convolutional structures, enabling efficient processing of long time series. In contrast, QRNN reduces computational costs by parallelizing operations and an optimized architecture, offering a significant advantage over traditional recurrent networks such as GRU and LSTM. The uniqueness of this approach lies in integrating these technologies, a combination that has not been previously applied in the context of time series forecasting.
2. Applying the results of modeling and experiments, we demonstrate that the TCN-QRNN model achieves the highest accuracy among analyzed models. The proposed model is 40% more accurate than LSTM and 8% more accurate than TCN-LSTM based on RMSE, MAE, and MAPE metrics, reducing evaluation time by 30%.
3. It was determined that the TCN-QRNN model has significantly fewer parameters compared to traditional recurrent neural networks like LSTM and GRU. The model is 20% smaller than LSTM and 8% smaller than GRU, in combination with a substantial increase in training speed, confirming the effectiveness of the proposed architecture in environments with limited computational resources.

The article is organized as follows: the "State-of-the-Arts" section describes a brief overview of current research and an analysis of existing approaches to power consumption forecasting. The section "Materials and Methods" explains the proposed architecture, including theoretical analysis of TCN and QRNN integration. In the "Modeling and Result" section, the datasets used are characterized, the rationale behind the chosen evaluation metrics is explained, and the modeling results on two datasets are presented along with comparisons to state-of-the-art methods. Finally, the "Conclusions" section summarizes the main findings and outlines prospects for future research.

State-of-the-arts

An analysis of modern literature on electricity forecasting indicates that recurrent neural networks, particularly LSTM and GRU models, are predominantly used for modeling. Some publications[12–20] focus solely on these models that highlight their popularity in time series forecasting tasks.

[12] explores the applicability of LSTM with various settings and optimizers to a hospital power consumption data. This article demonstrates the viability of LSTM to the power consumption forecasting task.

In paper[13] LSTM is compared to different machine learning approaches that includes ARIMA, different regression models, and ensemble of all the non-deep approaches. It covers training on all the countries provided by the ENTSO-E platform with monthly consumption report.

In study[14] a DWT-LSTM method is proposed for electricity load forecasting in Iran and Germany, considering the impact of cultural events and festivals. This method demonstrated the highest accuracy for forecasting over different time intervals, such as an hour, day, and year. A distinctive feature of this approach is its consideration of specific factors, such as the monthly calendar, that significantly increased accuracy during such events. The comparative analysis results confirmed the DWT-LSTM approach's superiority over existing methods.

Given the high accuracy of forecasting methods such as DWT-LSTM, their application in the port industry is important for planning and analyzing energy consumption. In[15], the study forecasts that electricity consumption in one of the big ports electricity consumption outgrows its throughput capacity. This highlights the need to secure additional power supply capabilities to ensure the port infrastructure's stable and efficient operation in the long term.

Including everything above, accurate modeling and forecasting of electricity consumption remains a challenging task due to many statistical characteristics of time series. As noted in the paper[16], econometric models and machine learning models were used to analyze power consumption during different stages of the COVID-19 lockdown. The results showed that climate change factors, like temperature, rainfall, average pressure, and wind speed, significantly influence power consumption. The most effective models were the autoregressive fractionally integrated moving average model and autoregressive Markov switching models with three states.

In paper[17] a deep learning structure that combines the LSTM and GRU models is proposed for energy consumption forecasting, considering weather variables and dependency on renewable energy sources. An enhanced sine-cosine swarm algorithm optimized the model's hyperparameters, significantly improving forecast accuracy. The proposed method demonstrates high efficiency for forecasting electricity generation by solar and wind stations and grid load, confirmed by low MSE values for both models.

Continuing the review of modern approaches to forecasting, in a study[18] proposes an approach that combines empirical mode decomposition with LSTM networks to forecast electricity demand accurately. This method efficiently decomposes time series into multiple intrinsic mode functions and residual components that allow more accurate energy load forecasting, including seasonal, daily, and hourly demand variances. To demonstrate the effectiveness of the approach, data on electricity consumption in the city of Chandigarh were used. Results showed that combining EMD with LSTM outperforms traditional methods, particularly recurrent neural networks, in forecasting electricity demand across different time intervals. Therefore, when investigating additional data transformation methods such as EMD or DWT, as shown in[14,18], LSTM demonstrates the highest forecasting accuracy. These studies include comparisons with other machine learning methods, confirming the superiority of LSTM in handling complex time series and enhancing the accuracy of electricity load predictions.

In subsequent studies, such as[19] emphasizes the importance of forecasting electricity load for effective design and development of energy systems. Forecasting is critical in increasing efficiency, ensuring the reliability and stability of energy systems, and reducing operating costs. In the scope of this study, different models were developed based on deep learning algorithms, especially LSTM and GRU, to ensure high forecasting accuracy of electricity load. It was found that GRU outperforms other models in terms of accuracy.

In the next investigation[20], which focuses on forecasting short-term electricity consumption in the context of smart grid development. Three learning algorithms were compared: LSTM, GRU, and Drop-GRU. The comparison was performed using power consumption data from French cities. The results show that LSTM has the highest accuracy and minimal forecasting errors, ensuring effective management of energy consumption and resource planning.

If an analysis is related to neural networks, LSTM and GRU are the most common models, as shown in studies[19,20]. To increase the forecasting accuracy in studies[21–25] architectures incorporating CNN and recurrent models like LSTM or GRU are widely used. Data preprocessing with convolutional layers effectively highlights special features, while recurrent models handle temporal dependencies in data.

The electric energy consumption prediction (EECP) is an important aspect of smart energy management systems, and its accuracy is critical to developing national energy policies. Recent studies have leveraged deep learning models to increase the accuracy of EECP. For example, the proposed EECP-CBL[21] model combines CNN to extract features and bidirectional LSTM (Bi-LSTM) to analyze time series. This model outperforms contemporary approaches in power consumption forecasting over various intervals, including long-term, medium-term, and short-term forecasting. Similarly, in the study[22] CNN is combined with GRU and enhanced with an attention mechanism to forecast energy consumption in optimizing large storage for smart systems. CNN extracts spatial features, while GRU effectively mitigates exploding gradient problems for long-term forecasting, demonstrating superior performance compared to other models.

In addition, hybrid models that combine CNN and GRU, as shown in[23], offer advantages in forecasting power consumption for buildings, demonstrating improved accuracy compared to traditional methods. These hybrid structures effectively capture spatial and temporal features, which are crucial for forecasting highly variable and uncertain power consumption in buildings. In another study[24], a proposed hybrid CNN-LSTM model achieved substantially better results in forecasting load on separate households than LSTM models. This approach effectively combines CNN for feature extraction from input data and LSTM layers for learning temporal dependencies in time series, leading to enhanced performance across various time horizons.

Emphasizes[25] the use of the CNN-LSTM model for forecasting energy consumption in residential households, where CNN layers extract complex spatial features, and LSTM layers model temporal dependencies in the components of time series. This combination shows almost perfect forecasting and reduction of RMSE compared to traditional methods. These researches emphasize the effectiveness of integrating CNN and LSTM or GRU to improve the accuracy of electricity consumption forecasting, particularly in the context of irregular trends and uncertainty in consumption data.

In the study[26], a TCN model is proposed that is specifically designed for time series processing. Its effectiveness is demonstrated in[27,28] where TCN is applied along LSTM and GRU models. The results showed that the proposed approach outperforms forecasts generated by state-of-the-art recurrent neural networks regarding accuracy.

In the study[29], a QRNN is proposed for power consumption forecasting, which achieves accuracy comparable to traditional methods but significantly outperforms them in computational speed. However, no studies have yet compared the performance of the TCN-QRNN model in forecasting tasks. The main goal behind developing QRNN was to improve existing LSTM models by enabling parallel computations. In 2024, a modified LSTM incorporating additional variables was proposed, substantially enhancing forecast accuracy[30]. However, the impact of the change on model speed was out of the scope of the study. Increasing the complexity of LSTM blocks naturally leads to a reduction in model speed.

Table 1 provides a comparative summary of the deep learning models applied in the reviewed studies, emphasizing their main contributions and limitations in the context of electricity consumption forecasting.

This emphasizes the importance of seeking efficient approaches to improve forecasting speed. In particular, integrating and combining various models opens up new opportunities for improving performance and forecasting accuracy in energy consumption-related problems.

Materials and methods

Based on literature analysis, LSTM and GRU were chosen as baseline models. Besides that, we examined the effect of adding TCN layers to these baseline models, comparing them with the proposed model. The proposed model is based on QRNN, as this recurrent layer leverages convolutional operations to accelerate the training process and computational efficiency. The forecasting horizon is set to 24 h. Discussed models include LSTM, GRU, TCN, and QRNN, each with unique architecture and mechanisms. Details on these models are provided below.

Long short-term memory model

LSTM is a recurrent neural network that effectively stores and processes long sequences of data. This allows LSTM networks to address long-term dependencies, an important problem in time series handling. LSTM's internal structure helps mitigate the issues of exploding or vanishing gradients, common in traditional recurrent neural networks, using specialized components that control data flow.

The LSTM model consists of multiple gates, each of which performs a special function during data processing:

- i_t – The input gate receives and integrates new information into the current memory state. It allows the network to determine which new data to remember.
- f_t – The forget gate determines which portion of past data should be discarded, thus regulating the unnecessary information for further processing.

References	Deep Models considered	Achievements	Limitations
[13]	LSTM	Testing on large amount of datasets; Many ML approaches considered	Limited dataset size; Only LSTM considered
[12,14]		Detailed analysis on hospital power data; DWT-LSTM method considering cultural/festival factors with high accuracy	Only LSTM considered; limited ML comparisons
		Forecast of port power consumption growth and capacity challenges	Single model; no extensive model analysis
[15]		Deep analysis of Busan port power consumption	Only one model considered, no model analysis
[16]			
[17]	LSTM, GRU	Combined LSTM and GRU optimized by sine-cosine algorithm; high accuracy on solar/wind data	Forecast horizon 1–3 steps
[18]	LSTM, RNN	EMD + LSTM approach for decomposed time series, improved accuracy	Single model explored
[19]	LSTM, GRU, RNN	Variable analysis; GRU outperforms others	No cross-validation; only recurrent models considered
[20]	LSTM, GRU, Drop-GRU	Comparison on French city data; LSTM best accuracy	Limited regularization techniques explored
[21]	CNN + BiLSTM	Hybrid CNN-BiLSTM model for multi-horizon forecasting	Variability in forecasting horizons across datasets
[22]	CNN + GRU, attention	CNN-GRU-attention for smart system energy forecasting	N/A
[23]	CNN + LSTM, CNN + GRU	Hybrid CNN-RNN models for building power consumption forecasting	Some models not fully described; lack of detailed metrics
[24]	CNN-LSTM	Hybrid CNN-LSTM outperforming standalone LSTM for household load	Short-term forecasting focus; GRU not evaluated
[25]		CNN-LSTM achieving significant RMSE reduction in residential data	N/A
[26]	TCN	Proposed TCN architecture for time series	N/A
[27]	TCN-GRU, TCN, LSTM, GRU	A hybrid TCN-GRU with Attention is explored.	Lacks information about reference model size and dataset size; TCN-LSTM model is not considered.
[28]	TCN, LSTM, ConvLSTM, GRU	An application of attention mechanism with TCN is explored across different tasks	Lacks computing efficiency analysis.
[29]	QRNN, LSTM	QRNN architecture proposed for power consumption forecasting and compared to LSTM	LSTM demonstrated slightly higher average accuracy (~1%)

Table 1. Summary of deep learning approaches for power consumption forecasting.

- o_t – The output gate is responsible for selecting part of the current state passed as an output from the network.
- c_t – the candidate cell state, which is used to compute new memory content for the cell update. It introduces potential new information to be incorporated into the memory.
- h_t – The model's output value produces the final network output based on the current memory state and ongoing processing.

The above structure allows LSTM to capture long-term data dependencies, preserve valuable information, and discard irrelevant data. This makes it effective for time series forecasting and processing problems. Next, we present Eqs. (1)-(6), which, displayed later, explain calculations for each gate and update the state of the LSTM model.

$$i_t = \sigma\ (W_i x_t + U_i h_{t-1} + b_i) \tag{1}$$

$$f_t = \sigma\ (W_f x_t + U_f h_{t-1} + b_f) \tag{2}$$

$$o_t = \sigma\ (W_o x_t + U_o h_{t-1} + b_o) \tag{3}$$

$$\widehat{c_t} = \sigma\ (W_c x_t + U_c h_{t-1} + b_c) \tag{4}$$

$$c_t = f_t \odot\ c_{t-1} + i_t \odot\ \widehat{c_t} \tag{5}$$

$$h_t = o_t \odot\ \sigma\ (c_t) \tag{6}$$

where:

- σ — sigmoid activation function;
- $W_i,\ U_i,\ b_i$ — weights and bias to be trained;
- x_t — input features on the current step;
- $\odot$ — elementwise multiplication.

These equations describe processes that occur at every LSTM training step, where gate values control data flow and help the model preserve long-term dependencies. This architecture enables LSTM to effectively handle long sequences that traditional RNNs fail to process due to exploding or vanishing gradients.

Gated recurrent unit model

GRU is a simplified version of LSTM created to solve the same problem with long-term dependencies in recurrent networks like exploding or vanishing gradients. In contrast to LSTM, GRU contains fewer parameters and uses only two gates: the update gate z_t and reset gate r_t (as displayed in Fig. 1[31]. It allows the reduction of model complexity, which makes it effective in situations where computation speed and optimization are critical. Equations (7)-(10) explain the main components of GRU.

$$z_t = \sigma\ (W_z x_t + U_z h_{t-1} + b_z) \tag{7}$$

$$\mathrm{r_t} = \sigma\ (\mathrm{W_r x_t} + \mathrm{U_r h_{t-1}} + \mathrm{b_r}) \tag{8}$$

$$\widehat{\mathrm{h_t}} = \varphi\ (\mathrm{W_h x_t} + \mathrm{U_h}\ (\mathrm{r_t} \odot\ \mathrm{h_{t-1}}) + \mathrm{b_h}) \tag{9}$$

$$\mathrm{h_t} = (1 - \mathrm{z_t}) \odot\ \mathrm{h_{t-1}} + \mathrm{z_t} \odot\ \widehat{\mathrm{h_t}} \tag{10}$$

where ϕ — hyperbolic tangent that adds nonlinearity to the model.

GRU is also known for its flexibility in adapting to various data types, which makes it suitable for time series forecasting, text processing, and other tasks involving sequential data. Its reduced number of parameters enables

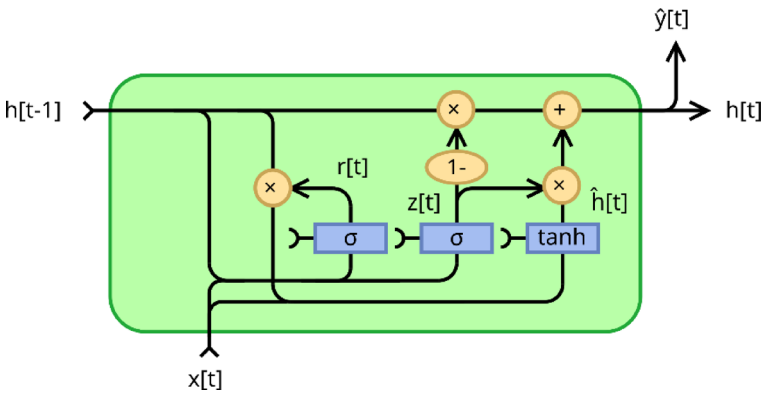


Fig. 1. GRU model schema[31].

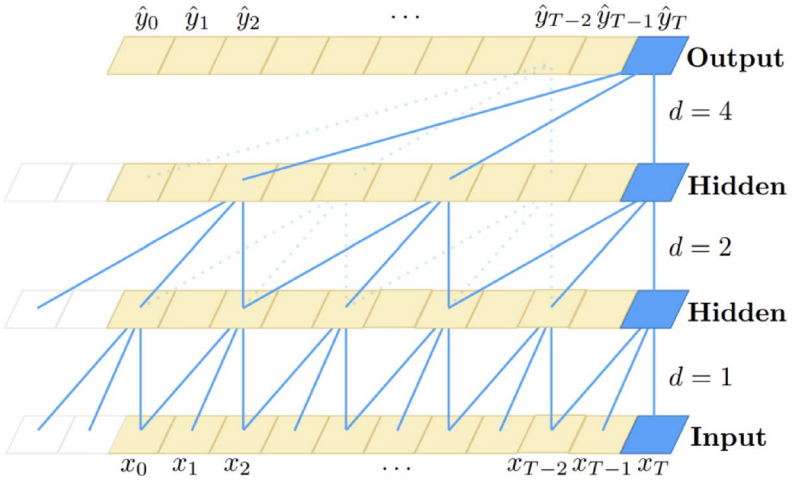


Fig. 2. Visualization of dilation with sizes (1, 2, 4).

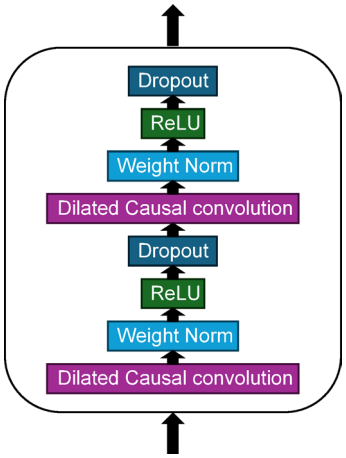


Fig. 3. Visualization of a TCN block.

faster training and lower computational overhead, making it advantageous for real-time applications and large-scale data processing.

Temporal convolutional network model

TCN is an architecture that applies extended convolutional layers for efficient time series processing instead of recurrent computations. The key feature of TCN is the use of dilated convolutions to ensure a wide input context without the need for an increasing number of parameters. This allows the model to capture long temporal dependencies accurately.

Another important component of the TCN architecture is its residual blocks, which include dropout, ReLU, and weight normalization. Such components contribute to training stability and model efficiency overall. However, TCNs are not prone to vanishing or exploding gradient problems commonly encountered in traditional recurrent networks, enabling them to handle sequences of arbitrary length. Figure 3 displays the structure of a TCN block.

Figure 2[26] illustrates the concept of dilation with sizes (1, 2, 4). It demonstrates how each subsequent layer covers a broader context without significantly increasing computation complexity.

As illustrated in Fig. 3, the TCN block does not contain any recurrent operations. This architectural property allows the entire input sequence to be processed in parallel at each layer, significantly improving computational efficiency compared to recurrent models.

Due to its modular architecture, TCN provides high flexibility in addressing time series analysis problems. Using dilated convolutions combined with residual blocks allows the model to scale for complex tasks while preserving computational efficiency and stability.

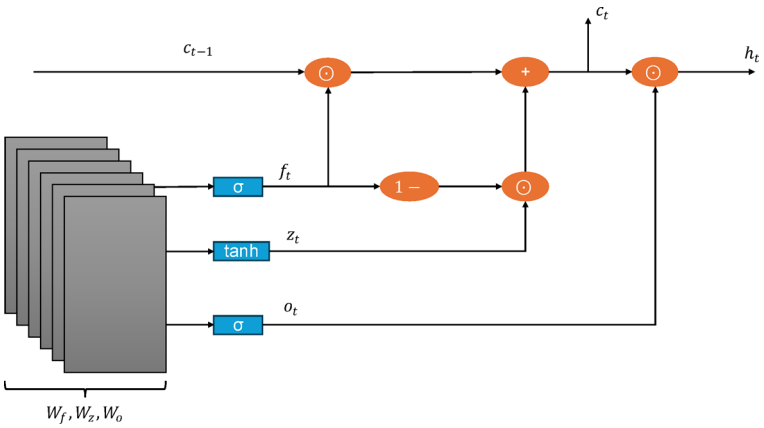


Fig. 4. Visualization of a QRNN block.

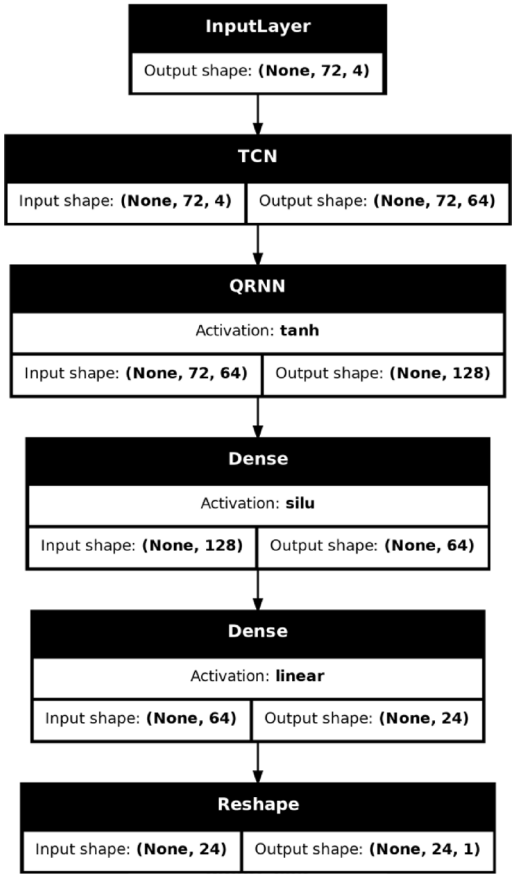


Fig. 5. Proposed model visualization.

Quasi-recurrent neural network

QRNN is an architecture combining convolutional operations with recurrent mechanisms, effectively balancing computational speed and the ability to model sequence dependencies. The main distinction between QRNN and traditional recurrent networks is the use of convolutional operations for input processing instead of relying solely on recurrent computations. This significantly reduces computational overhead and accelerates the training process. Figure 4 visualizes the structure of the QRNN block as explained below.

A QRNN block employs a two-stage processing scheme, analogous to the convolutional and pooling layers, respectively. Equations (11)-(13) describe the application of convolutional filters to the input. Here, c_t is responsible for computing convolution over input data x_t, which defines a context for the next state. Filter f_t

has the role of a forget gate, regulating which portion of data to discard. Output gate o_t controls which part of the memory state s_t are used to generate the current output.

Then, Eqs. (14)-(15) describe the recurrent pooling stage, which computes the output from the block. The general memory state s_t is defined as a combination of the previous state c_{t-1} and new context z_t, whose weights are regulated by forget gate f_t. Finally, output h_t is computed by elementwise multiplication of memory state s_t and output gate o_t.

$$c_t = \varphi\ (W_c * x_t) \tag{11}$$

$$f_t = \sigma\ (W_f * x_t) \tag{12}$$

$$o_t = \sigma\ (W_o * x_t) \tag{13}$$

$$s_t = f_t \odot\ c_{t-1} + (1 - f_t) \odot\ z_t \tag{14}$$

$$h_t = o_t \odot\ s_t \tag{15}$$

QRNN offers significant computational speed advantages due to its ability to process the whole input sequence in the first stage and limit the recurrent component of the block. In addition, this architecture allows effective handling of long sequences to mitigate problems with exploding or vanishing gradients and adapt to complex issues. Consequently, QRNN is well-suited for applications that require accuracy and speed, like time series analysis, natural language processing, or speech recognition.

Architecture of the proposed model

The proposed model is designed with modern architectural decisions to ensure effective time series analysis and forecasting. It consists of multiple layers that balance computational speed, the ability to handle long-term dependencies, and forecasting accuracy. The key components are described below.

At the first stage, the model uses a TCN block, which provides parallel temporal feature extraction through dilated convolutions, capturing hierarchical temporal patterns at different levels.

The second component is a QRNN block. It contributes selective information gating, enabling the model to retain or discard temporal information. As noted in[29], the QRNN cannot produce the result on its own; therefore, an additional preprocessing block is required.

At the next stage, a fully connected layer with the SiLU activation function is applied. This function is chosen over the traditional ReLU based on findings in[32]. SiLU provides smoother activation, promoting better gradient consistency and stability during training. Consequently, the model achieves higher accuracy, especially in problems requiring complex non-linear dependency modeling.

Subsequently, the model incorporates another fully connected layer that produces an actual forecast for the specified horizon of 24 h. This layer is the final computational step.

The last step of the model reshapes the output into a format suitable for analysis or visualization. This step ensures alignment between the model's output and problem requirements, making the model applicable to a wide range of applied scenarios.

Other TCN-* series models share the same structure but may incorporate different variations of TCN or recurrent blocks depending on experimental settings. This approach facilitates the comparison of various configurations and enables the optimization of the architecture for specific conditions.

Figure 5 presents a visualization of the proposed model, demonstrating the sequence of layers and general structure.

The proposed TCN-QRNN architecture combines the advantages of convolutional operations for processing temporal features with the efficiency of QRNN for sequence modeling, resulting in improved forecasting accuracy and speed compared to existing approaches.

	Value	feels_like	pressure	wind_speed	temp	temp_min	temp_max	wind_gust	wind_deg	rain_1h	snow_1h
Value	1.000000	-0.098018	-0.004815	0.128988	-0.080825	-0.079770	-0.084195	0.025719	0.007692	-0.040763	0.012752
feels_like	-0.098018	1.000000	-0.068652	-0.128380	0.992790	0.990748	0.991383	0.037227	0.038445	0.096371	-0.048240
pressure	-0.004815	-0.068652	1.000000	-0.274603	-0.087221	-0.089662	-0.081235	0.102700	-0.107691	-0.140857	-0.242022
wind_speed	0.128988	-0.128380	-0.274603	1.000000	-0.058265	-0.052593	-0.066186	0.630654	0.116852	0.011376	0.271108
temp	-0.080825	0.992790	-0.087221	-0.058265	1.000000	0.998234	0.998021	0.078717	0.036899	0.096838	0.106201
temp_min	-0.079770	0.990748	-0.089662	-0.052593	0.998234	1.000000	0.995861	0.082739	0.039138	0.100499	0.115664
temp_max	-0.084195	0.991383	-0.081235	-0.066186	0.998021	0.995861	1.000000	0.077289	0.034862	0.100271	0.107872
wind_gust	0.025719	0.037227	0.102700	0.630654	0.078717	0.082739	0.077289	1.000000	0.090951	0.000246	-0.157583
wind_deg	0.007692	0.038445	-0.107691	0.116852	0.036899	0.039138	0.034862	0.090951	1.000000	-0.053238	0.024683
rain_1h	-0.040763	0.096371	-0.140857	0.011376	0.096838	0.100499	0.100271	0.000246	-0.053238	1.000000	NaN
snow_1h	0.012752	-0.048240	-0.242022	0.271108	0.106201	0.115664	0.107872	-0.157583	0.024683	NaN	1.000000

Fig. 6. Correlation matrix among variables from the first dataset.

Modeling and results

This section provides a detailed description of the datasets utilized to evaluate the proposed method's performance, specifying the metrics employed to assess its effectiveness and showcasing the results obtained from the implemented approach.

Datasets

Two datasets were utilized in the scope of this study. Each dataset is described below. The first dataset[33] contains hourly power consumption data in Luxembourg for 2006–2015 obtained from the website https://www.entsoe.eu/. In addition, historical weather data were sourced from https://openweathermap.org/. The total dataset size is 87,638 records. Power consumption data are stored in Excel, and hourly values are provided daily. Two missing records were identified and excluded from the analysis. The weather dataset, which is provided in CSV format, does not have any absent values. Both datasets were merged based on a common date parameter.

The selected features based on the correlation analysis are presented in Table 2.

The weather data indicate a typical oceanic-continental climate: it is mild with wet winters and moderately warm.

These variables were selected because they showed the highest correlation with the target variables, ensuring their relevance as input parameters.

It should be noted that variables feels_like, temp, temp_min and temp_max represent the same parameter – the outdoor temperature. Therefore, only one variable with the highest correlation is selected. In Fig. 6, presented below, the correlation matrix can be found.

Each variable is min-max normalized according to the Eq. (16):

$$x = \frac{x - x_{min}}{x_{max} - x_{min}} \tag{16}$$

Regarding the stationary of the target variable, the detailed analysis demonstrated that it is stationary. To verify this, two tests were performed:

1. Dickey-Fuller test;
2. KPSS test.

According to the KPSS test results, The test statistic is 0.28, p-value – 0.1. Critical values for different levels of significance:

- 10%: 0.347;
- 5%: 0.463;
- 2.5%: 0.574;
- 1%: 0.739.

Dickey-Fuller test results are −8.5, p-value – 0. Critical values for different levels of significance:

- 1%: −3.430;
- 5%: −2.862;
- 10%: −2.567.

Results of both tests confirm that data do not exhibit an upward trend and additional transformations of target variables are unnecessary. Stationarity indicates that average, variance, and autocorrelation are constant. It also mitigates false correlations based on common trends.

The dataset exhibits a pronounced daily seasonality, as illustrated in Fig. 7.

The second dataset is power consumption in London[34]. It contains a subset of smart meter data collected from November 2011 to February 2014. This dataset includes hourly and half-hourly values of power consumption. Data is stored in CSV format, featuring household identification code, consumption value, and historical weather data obtained from the DarkSky platform.

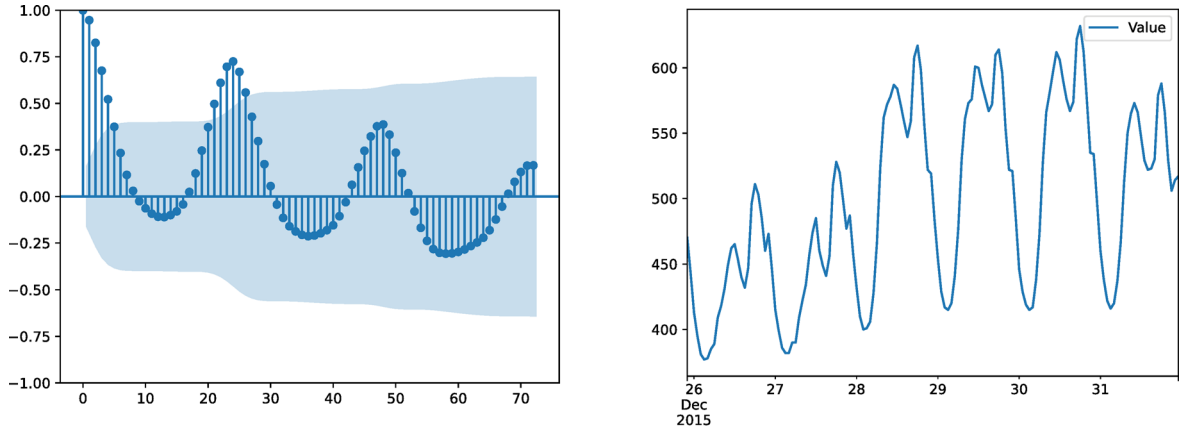
The selected features are described in Table 3.

Compared to the Luxembourg dataset, the weather in London is slightly warmer and more humid, which can be attributed to its insular location on the British Isles, whereas Luxembourg is situated in continental Europe.

The use of two geographically and contextually distinct datasets – national-scale data from Luxembourg and household-level smart meter data from London – allows us to evaluate the model's generalizability across diverse environments and usage patterns.

Variable	Units	Description	min	max	25% quantile	75% quantile
Value	Megawhatts	historical data about power consumption	148	1188	648	831
feels_like	Degrees Celsius	Apparent temperature	−21.85	40.94	1.63	14.76
wind speed	Meters/second	Wind speed	0.02	29.3	2.06	4.33
pressure	Hectopascals	Atmospheric pressure	968	1044	1011	1022

Table 2. Description of features in the Luxembourg dataset.



a) Autocorrelation for the last three days

b) Visualization of power consumption for 5 days

Fig. 7. Seasonality of the target variable.

Variable	Units	Description	min	max	25% quantile	75% quantile
Value	kWatt/hh	Power consumption	0.014	3.516	0.07	0.487
Temperature	Degrees Celsius	Actual outdoor temperature	−5.64	32.42	6.4	14.3
Humidity	%	Humidity	0.23	1	0.7	0.89
Pressure	hectopascals	Atmospheric pressure	976	1040	1010	1020

Table 3. Description of features in the London dataset.

Similarly to the first dataset, ADF and KPSS tests were conducted to verify the stationarity of the target variables. The results of the ADF test were as follows:

- Statistic: −16;
- p-value: 0.

Critical values:

- 10%: 0.3;
- 5%: 0.46;
- 2.5%: 0.57;
- 1%: 0.7.

According to the test, the sequence is stationary.

The results of the KPSS test are different:

- Statistic: 14;
- p-value: 0.01.

Critical value:

- 10%: 0.347;
- 5%: 0.463;
- 2.5%: 0.574;
- 1%: 0.739.

According to the KPSS test results, the series is not stationary. The difference between those tests indicates that there is a deterministic trend in the series. The pairwise difference method is applied to remove non-stationarity.

In the scope of the study, neural networks will be applied using both inputs of pairwise difference and actual values, as neural networks can train on non-stationary data.

In addition to the stationarity analysis, the dataset also demonstrates clear seasonality patterns. Autocorrelation visualization is presented in Fig. 8, and power consumption for 300 h is presented in Fig. 9.

As observed, power consumption has pronounced seasonality. Every variable in this dataset is min-max normalized according to the Eq. (16).

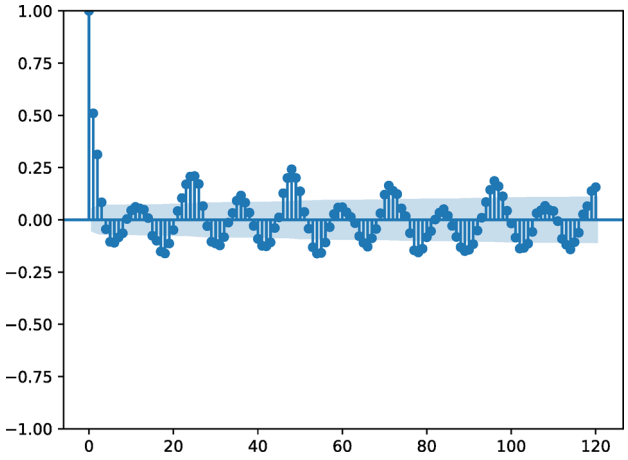


Fig. 8. Autocorrelation visualization for the second dataset.

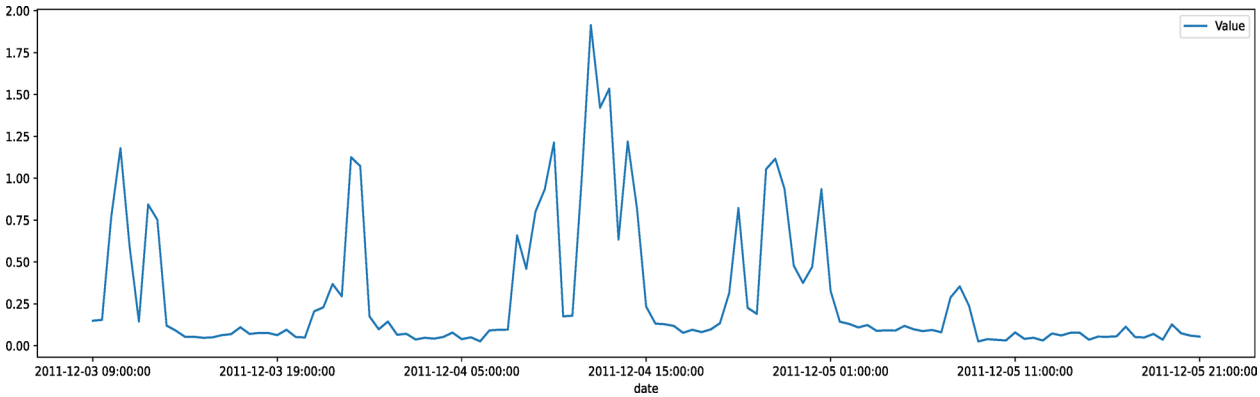


Fig. 9. Visualization of power consumption over a 5-day period for the London dataset.

Indicators of performance

The effectiveness of the proposed approach was evaluated using multiple performance metrics. The optimization process is based on the MSE loss function, which minimizes the difference between the actual and predicted values, with greater emphasis on larger deviations.

In addition to the loss function, model effectiveness is also assessed using RMSE, MAE, and MAPE metrics. RMSE estimates the average magnitude of the forecasting error, giving more weight to larger differences. MAE represents the average of absolute differences between actual and predicted values, regardless of the error direction. MAPE expresses the error as a percentage, which allows easier comparison of model accuracy across different data scales.

Beyond the specified accuracy metrics, additional aspects are considered to provide a comprehensive assessment of model performance. These include model size, forecasting time on the test set, single-step evaluation time, and training time per batch.

The metrics described above are defined in Eqs. (17)–(20):

$$MSE = \frac{1}{N}\sum_{t=1}^{N}(\boldsymbol{y_t} - \widehat{\boldsymbol{y}}_{\boldsymbol{t}})^2 \tag{17}$$

$$RMSE = \sqrt{\frac{\sum_{t=1}^{N}\left(y_t - \widehat{y_t}\right)^2}{N}} \tag{18}$$

$$MAPE = 100\frac{1}{N}\sum_{i=1}^{N}\left|\frac{y_t - \widehat{y_t}}{y_t}\right| \tag{19}$$

$$MAE = \frac{\sum_{t=1}^{N}\left|y_t - \widehat{y_t}\right|}{N} \tag{20}$$

where y_t – actual values of the target variable, $\widehat{y_t}$ – predicted values of the target values, N – number of elements.

Model	Luxembourg			London		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE
TCN (3, [1, 3, 5])-QRNN (128)	0.0065	0.0022	0.42	0.0220	0.0069	23

Table 4. Accuracy results for TCN-QRNN model on the two datasets.

Model	Luxembourg			London		
	Test set eval time (s)	Av. train step (ms)	Av. test step (ms)	Test set eval time (s)	Av. train step (ms)	Av. test step (ms)
TCN (3, [1, 3, 5])-QRNN (128)	14	8.8	3	15	9.2	3.2

Table 5. Efficiency of the TCN-QRNN model on the two datasets.

Experiment setup

Model training was performed using an NVIDIA T4 GPU on Google Colab. The computing efficiency of the models was evaluated on a Raspberry Pi 4 Model B and a laptop equipped with an Intel Core Ultra 9 processor and an NVIDIA GeForce RTX 4070 GPU (8 GB). Keras with Tensorflow backend was used for code development. Computational resources were chosen to enable efficient testing on large datasets while maintaining acceptable training and evaluation times.

The main accuracy metrics described by Eqs. (17)-(20) are used to analyze the model's performance: MSE, RMSE, MAPE and MAE. In addition to accuracy, the processing speed on the test set was assessed based on the average evaluation time and average inference step duration. Training was conducted using four training/validation splits: 60/40, 70/30, 80/20, and 90/10. The training procedure is summarized in Listing 1 below.

The last 15% of every dataset is selected as a test set for model assessment, while the other 85% is used as a training and validation set. Rolling-window cross-validation is used to ensure the reliability of the obtained results.

Model estimation on a test set was performed after each full training phase. Each model was trained for 25 epochs using the Adam[35] optimizer. The initial learning rate is 5×10^{-4} and automatically decreases if there is no increase in validation loss for two consecutive epochs. Hyperparameter search was performed using the Optuna[36] library, which provided additional flexibility in model tuning. The training procedure is summarized in Listing 1 below.

```
for each (train_data, validation_data) in dataset_splits:
  for epoch = 1 to 25 do
    train the model using train_data
    validate the model using validation_data
    if validation accuracy does not improve for 2 consecutive epochs then
      reduce learning rate by factor of 0.1
  end for
  evaluate the model on test_data
  if evaluation accuracy > best recorded accuracy then
    save the model
end for
```

Listing 1. Pseudocode for model training and evaluation.

The training process follows a rolling-window cross-validation strategy using four different train/validation splits. As shown in Listing 1, each model is trained for 25 epochs. If the validation accuracy does not improve for two consecutive epochs, the learning rate is reduced by a factor of 0.1. After each training session, the model is evaluated on the test set. The best-performing model based on evaluation accuracy is saved for final testing.

All models were developed to forecast energy consumption over a 24-hour horizon using the preceding 144 h as input data. This approach enhances accuracy and ensures model stability with respect to fluctuations in input variables.

For recurrent models (LSTM, GRU, and QRNN), hidden sizes of 64 and 128 were explored. For the GRU and LSTM models, one and two stacked layers were evaluated.

For the TCN model, kernel sizes of 3 and 6 were used, along with dilation rates of[1,1,3,3,5,5,8]. These dilation configurations allow the model to capture dependencies across varying, non-overlapping temporal scales.

Models	RMSE	MAE	MAPE
Residual Dilated LSTM + ES [13]	–	–	2
GRU (128–128)	0.011	0.0077	1.39
LSTM (128–128)	0.0102	0.0064	1.2
TCN (3, [1, 3, 5])-GRU (128)	0.0067	0.0025	0.47
TCN (3, [1, 3, 5])-LSTM (128)	0.0066	0.0024	0.46
TCN (3, [1, 3, 5])-QRNN (128)	**0.0065**	**0.0022**	**0.42**

Table 6. Forecasting accuracy on a Luxembourg power consumption dataset.

Models	Test set eval time, s	Av. test step, ms	Av. predict. time ms
GRU (128–128)	22	17	39
LSTM (128–128)	25	18	49
TCN (3, [1, 3, 5])-GRU (128)	22	15	40
TCN (3, [1, 3, 5])-LSTM (128)	24	15	36
TCN (3, [1, 3, 5])-QRNN (128)	**14**	**10**	**33**

Table 7. Efficiency comparison of the Luxembourg dataset on a test laptop.

Models	Test set eval time, s	Av. test step, ms	Av. predict. time ms
GRU (128–128)	68.8	49.8	144
LSTM (128–128)	93.6	78.2	156
TCN (3, [1, 3, 5])-GRU (128)	53.4	44.2	129
TCN (3, [1, 3, 5])-LSTM (128)	51.2	42.2	130
TCN (3, [1, 3, 5])-QRNN (128)	**32.4**	**27**	**113**

Table 8. Efficiency comparison of the Luxembourg dataset on a raspberry Pi.

Results

Table 4 displays the accuracy metrics for the best setup of the proposed model. Table 5 presents computational efficiency of the model.

The results demonstrate the high accuracy of the proposed TCN-QRNN model on both datasets, as confirmed by low RMSE, MAE, and MAPE values. Additionally, the model demonstrates high efficiency, ensuring rapid evaluation over the test set (14 and 15 s) and efficient test and train steps. The slight difference between the two datasets indicates the model's generalizability and its ability to adapt to varying energy consumption environments.

Discussion

In this section, we analyze the proposed TCN-QRNN model's performance compared to other models presented in Tables 3 and 5, and 6, focusing on forecasting accuracy, efficiency, and model size. The results indicate that replacing a recurrent block with a TCN block drastically improves prediction accuracy while maintaining or improving computational performance and reduces model size.

Table 6 also includes the MAPE reported in[13] which used the ENTSO-E dataset and provided forecasts for multiple countries, including Luxembourg. Although the model in[13] – a hybrid residual dilated LSTM with exponential smoothing – differs significantly in structure and training settings, the reported MAPE offers a useful benchmark for contextualizing the accuracy of our approach.

As we can see from the table above, the proposed model achieves the highest accuracy among all approaches. It is 40% more accurate than classical recurrent networks such as LSTM and 8% more accurate than TCN-LSTM.

Furthermore, the computational efficiency comparison presented in Tables 7 and 8 indicates that replacing the recurrent block with TCN does not affect processing time. At the same time, the proposed model achieves up to 44% faster evaluation time compared to LSTM and GRU models. To test the significance of observations we performed the ANOVA test for both testing devices. For every statistic we gather 40 records. The results are shown in Table 9 below.

Such a difference between RPi and a target laptop is explained by the computing capabilities: RPi has an ARM CPU 1.5 GHz while the latter has a capable NVIDIA GPU.

In addition, TCN allows reducing model size, leading to better optimization of computational resources: TCN-QRNN is 20% smaller than LSTM and 8% less than GRU.

Performance criteria	Raspberry Pi	Laptop
Test set eval time	1403.9	47.7
Av. test step	3821	42.1
Av. prediction time	357.4	51.6

Table 9. F-statistic values for target devices performance testing.

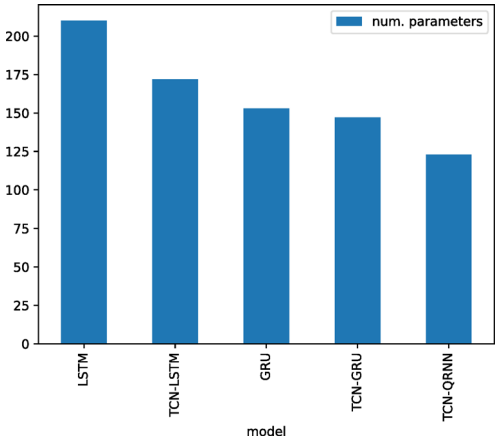


Fig. 10. Visualizing model sizes for performance evaluation.

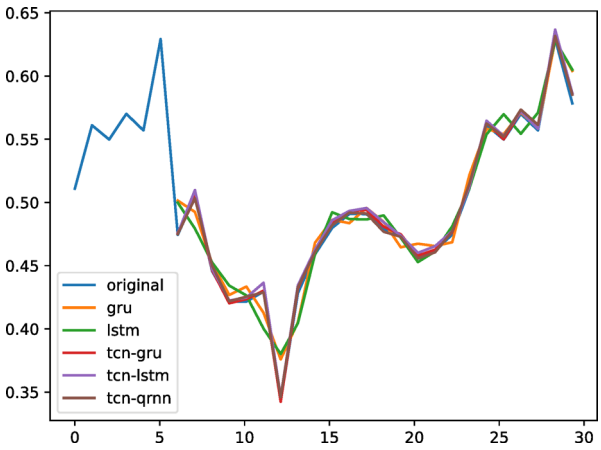


Fig. 11. Comparison of prediction results to the actual value for different model architectures on the first dataset.

Figure 10 presents model sizes in thousands of parameters, displaying the advantage of the proposed model by a large decrease in a number of parameters in comparison to other approaches.

Figure 11 visually compares the forecasting accuracy of considered models for the Luxembourg dataset.

During training on the second dataset (London Smart Meters), we used two variations of input data: weather data with elementwise differences in power consumption and weather data with actual consumption values. The latter approach shows better results.

Similarly, the model with the biggest recurrent size demonstrated the best performance for the London PC dataset and a kernel size 3, as presented in Table 10.

The second dataset's computational efficiency is almost the same as the results for the first dataset in Tables 7 and 8. As for the first dataset, Fig. 12 visualizes the comparison of actual and predicted values for the second dataset.

Additionaly, we explored relative metrics described by Eqs. (21)-(23). The results are provided in Tables 11 and 12.

Models	RMSE	MAE	MAPE
GRU (128–128)	0.0255	0.011	38.5
LSTM (128–128)	0.0339	0.0197	60
TCN (3, [1, 3, 5])-GRU (128)	0.0221	**0.0069**	23.5
TCN (3, [1, 3, 5])-LSTM (128)	0.0221	0.0072	24.25
TCN (3, [1, 3, 5])-QRNN (128)	**0.0220**	**0.0069**	**23**

Table 10. Visual comparison of predicted and actual values for different models on the Luxembourg dataset.

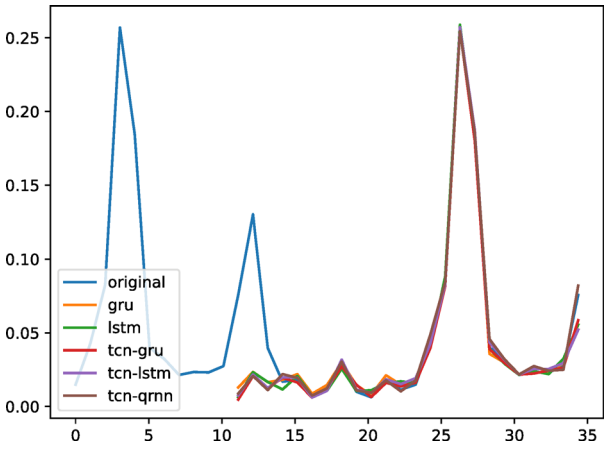


Fig. 12. Comparison of predicted and actual values for different model architectures on the London dataset.

Models	RMSRE	MARE	MSRE
GRU (128–128)	0.0233	0.0158	$5.47 \cdot 10^{-4}$
LSTM (128–128)	0.0245	0.0168	$6.03 \cdot 10^{-4}$
TCN (3, [1, 3, 5])-GRU (128)	0.0127	0.0044	$1.62 \cdot 10^{-4}$
TCN (3, [1, 3, 5])-LSTM (128)	0.0129	0.0043	$1.68 \cdot 10^{-4}$
TCN (3, [1, 3, 5])-QRNN (128)	**0.0126**	**0.0040**	$1.6 \cdot 10^{-4}$

Table 11. Additional evaluation metric for Luxembourg dataset.

Models	RMSRE	MARE	MSRE
GRU (128–128)	**1.06**	0.248	**1.14**
LSTM (128–128)	1.24	0.377	1.59
TCN (3, [1, 3, 5])-GRU (128)	1.10	0.231	1.25
TCN (3, [1, 3, 5])-LSTM (128)	**1.06**	0.236	1.16
TCN (3, [1, 3, 5])-QRNN (128)	**1.06**	**0.224**	1.16

Table 12. Additional metrics for London smart meters dataset.

$$RMSRE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \widehat{y_i}}{y_i}\right)^2} \tag{21}$$

$$MSRE = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \widehat{y_i}}{y_i}\right)^2 \tag{22}$$

$$MARE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \widehat{y_i}}{y_i}\right|^2 \tag{23}$$

An ablation study was also conducted by comparing the performance of TCN-GRU, TCN-LSTM, and TCN-QRNN architectures, clearly showing the individual contributions of the QRNN and TCN components to the overall improvement.

In summary, the proposed TCN-QRNN model demonstrates superior forecasting accuracy, higher efficiency, and a smaller model size than other discussed approaches. Replacing the recurrent layer with a TCN block significantly improves forecasting effectiveness without increasing computational cost, making the model more competitive for energy forecasting problems.

Conclusions

This paper analyzes the effectiveness of applying the TCN-QRNN model for forecasting power consumption using the Luxembourg power consumption dataset. The proposed model is compared with LSTM and GRU architectures, as well as their combinations with TCN blocks. The results demonstrate that TCN-QRNN outperforms the compared models in terms of forecasting accuracy, computational efficiency, and model compactness. In particular, the TCN-QRNN model achieves up to 44% faster evaluation time and improved RMSE, MAE, and MAPE metrics while requiring fewer parameters. These characteristics make the model a competitive solution for applications requiring fast and accurate predictions on large-scale time series data.

Future work may involve integrating enhanced versions of TCN with alternative convolutional architectures to improve performance in real-world environments further. To broaden its applicability, the TCN-QRNN model can also be adapted for non-linear or unevenly spaced time series data. Considering the limitations of classical LSTM models, particularly their growing computational demands with increased model size, developing hybrid architectures – such as QRNN – that combine parallelism with convolutional design remains a promising direction for achieving both high efficiency and accuracy.

Data availability

The datasets used in this study are publicly available. The first dataset, Luxembourg Power Consumption, can be accessed at[33] (https://www.kaggle.com/datasets/arihilmir/luxembourg-power-consumption). The second da taset, Smart Meters in London, is available at[34] (https://www.kaggle.com/datasets/jeanmidev/smart-meters-in-london).

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Author contributions

Conceptualization, L.M.; methodology, L.M.; software, R.L.; validation, R.L.; formal analysis, L.M.; investigation, L.M, R. L.; resources, R.L.; data curation, R.L.; writing—original draft preparation, L.M.; writing—review and editing L.M., R. L.; visualization, R.L.; supervision, L.M.; All authors have read and agreed to the published version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to L.M.

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